

Climate change refers to long-term shifts in average weather patterns. While the climate has always changed, today's changes are happening faster and are mostly caused by human activities.

Australia is already feeling the effects of climate change. These changes can impact our:

- farming businesses
- rural and regional communities
- natural environments.

You can manage risks to your business and community by understanding climate risks, managing emissions and building resilience.

How Australia's climate is changing

Australia's climate is already changing. Every two years the Bureau of Meteorology and CSIRO release a report called [State of the Climate](#) to explain current and future changes.

The 2024 report found that:

- Each decade since 1950 has been warmer than the one before it
 - Australia's climate has warmed by an average of 1.51 ± 0.23 °C since 1910.
 - Sea surface temperatures have risen by an average of 1.08 °C since 1900.
 - Extreme heat events over land and oceans are more frequent.
- Rainfall patterns are changing
 - South-west Australia has seen a 16% decrease in April–October rainfall since 1970.
 - South-east Australia has experienced a 9% decrease in April–October rainfall since 1994.
 - Heavy short-term rainfall events are becoming more intense.
 - Rainfall and streamflow have increased in parts of northern Australia since the 1970s.
- Weather conditions that increase the risk and severity of bushfires are more frequent
 - Extreme fire weather and longer fire seasons have increased across much of Australia since the 1950s.
- Sea levels are rising
 - Sea levels are rising around Australia, leading to more frequent extreme high tides and increased risk of coastal flooding and damage.

The [State of the Climate 2024 report](#) and the [National Climate Risk Assessment](#) (Australian Climate Service) also explain what is projected in the future.

How climate change affects Australian farms

You can explore these resources to understand how climate change may impact your farm.

My Climate View

[My Climate View](#) helps to see how climate change may impact your land in the next 50 years by showing future climate projections. It can provide seasonal forecasts for the upcoming 1-3 months as well as climate projections for the 2030s, 2050s and 2070s. This can help you make long-term strategic decisions about your enterprise.

Regional Climate Change Explorer

[Regional Climate Change Explorer](#) offers detailed information on how climate change could affect different places in Australia. It helps you understand the changes you could expect in your area.

Build resilience on farm

The shift in recent decades towards higher temperatures and lower winter rainfall has had significant effects on many farms ([ABARES](#)). However, many producers are using new technologies and improving land management practices to help increase farm productivity.

In 2021, [ABARES](#) found that broadacre farm productivity increased by 28%, while cropping farms saw a 68% boost over the last 30 years. These productivity gains have offset the negative effects of climate change, such that productivity levels have increased or at least remained stable over the last 30 years.

Looking ahead, it is important to continue building resilience by understanding climate risks, investing in new technology and innovation, and improving land management practices. This will help reduce the impacts of climate change on productivity and profitability.

Explore [resources and programs to help you build resilience](#) (DAFF). The state and territory resources listed below can also help you understand how to build resilience.

Additional resources

Explore further resources on climate change and agriculture.

State and territory resources

Each state and territory government offers resources to support understanding the impact of climate change on agriculture. These include practical advice for local conditions as well as broader strategies and reports.

Resources available in each state and territory:

- Australian Capital Territory: [Climate change](#)
- New South Wales: [Climate change impacts on our agriculture](#)
- Northern Territory: [NT Climate Change Response](#)
- Queensland: [Queensland Future Climate Dashboard](#)

- South Australia: [Climate change](#)
- Tasmania: [Climate change and agriculture](#)
- Victoria: [Agriculture and climate change](#)
- Western Australia: [Climate and weather](#)

See more

Learn more about climate change and farm resilience:

- [ABARES](#): Climate and drought data and insights.
- [ABARES](#): Research on how different industries and regions have been affected and are adapting.
- [ACS](#): Climate risks and natural hazards.
- [DAFF](#): Programs and services to help you prepare for, manage and recover from extreme events.
- [DCCEEW](#): Climate change and its effects.
- [Climate Change in Australia](#): Tools and reports on national climate information and projections.

To prepare your business for the long term, you may want to explore options to [manage your emissions](#).

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/climatechange/carbon-farming-outreach-program/hub/about-emissions-management-ag/climate-change>